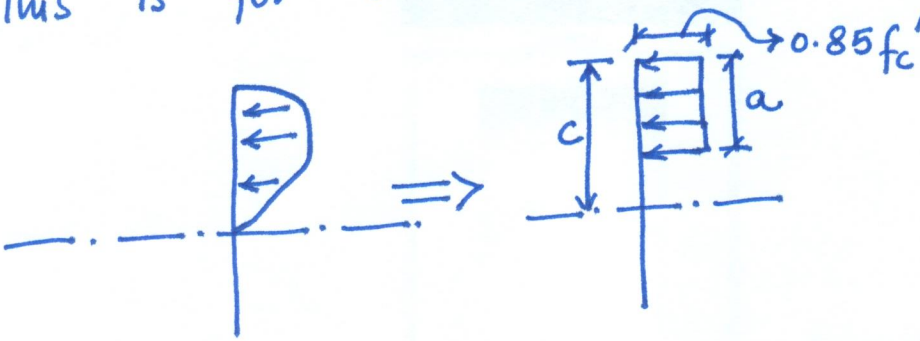
Equivalent Rectangular (Whitney's) Stress block

- This is for ultimate strength level only



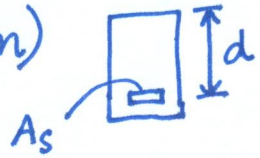
$a = \beta_1 c$

$$\begin{cases} \beta_1 = 0.85 & \text{for } f_c' \le 4000 \text{ psi} \\ = 1.05 - 0.05 f_c' & \text{for } 4000 \le f_c' \le 8000 \text{ psi} \\ = 0.65 & \text{for } f_c' > 8000 \text{ psi} \end{cases}$$

($\beta_1 = 1.09 - 0.008 f_c'$ in SI Units)

$0.85 \le \beta_1 \le 0.65$

Nominal Moment (singly-reinforced section):



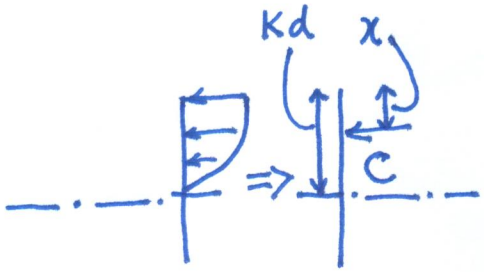
$$M_n = T\left(d - \frac{a}{2}\right) = A_s f_y \left(d - \frac{\beta_1 c}{2}\right)$$

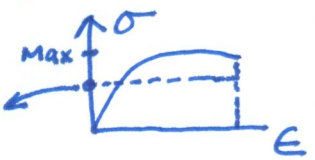
$$C = T \Rightarrow 0.85 f_c' a b = A_s f_y$$

$$\beta_1 c = a = \frac{A_s f_y}{0.85 f_c' b}$$

A General Equivalent Rectangular Stress block

Lets define three Stress-strain Model constants as follows;

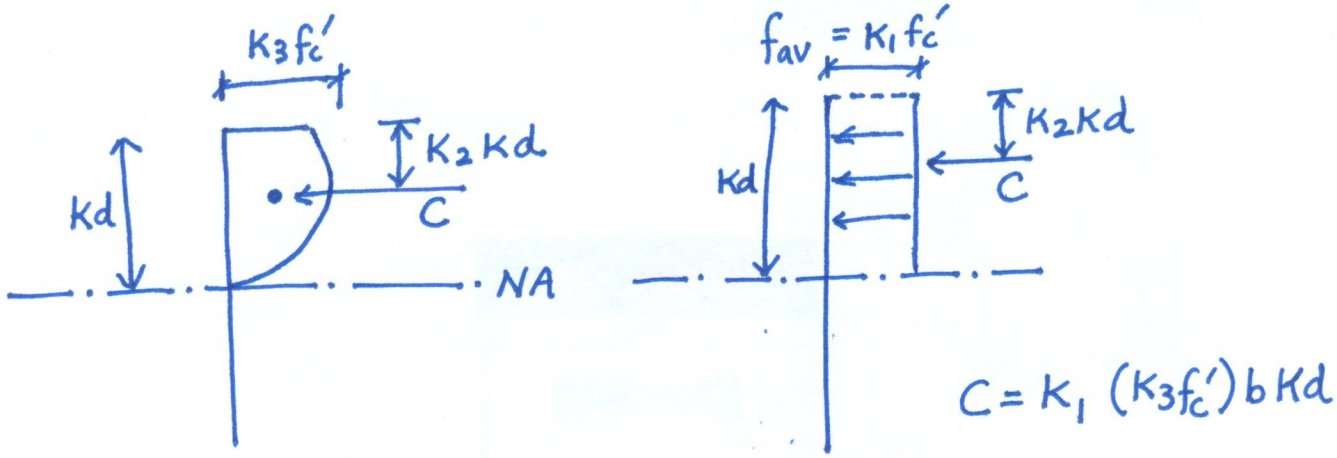




$$K_1 = \frac{\text{Average stress}}{\text{Max. Stress}}$$

$$K_2 = \frac{\text{Extreme Fiber to Resultant}}{\text{NA Depth}} = \frac{x}{Kd}$$

$$K_3 = \frac{\text{Max. Stress}}{\text{Cylinder strength}}$$



Nominal Moment:

$$M_n = A_s f_y (d - K_2 Kd)$$

$$C = T \Rightarrow A_s f_y = K_1 K_3 f_c' b Kd$$

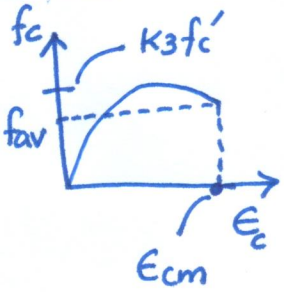
$$Kd = \frac{A_s f_y}{K_1 K_3 f_c' b}$$

This general equivalent stress block will use the properties of actual $\sigma\epsilon$ model (K_1, K_2, K_3) and can handle all stress states of cross-section (not only ultimate strength).

Now, the $\sigma\epsilon$ Model constants are the function of extreme fiber strain. They will be computed for all values of ϵ_{cm} (can) used in the analysis.

Determination of σε Model Constants

Condition for K_1 : $K_1 = \dfrac{f_{av}}{K_3 f_c'}$

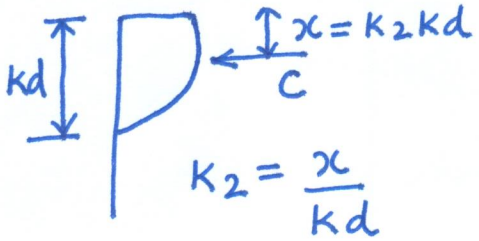


Area under actual σε Model = Area of equivalent rectangle

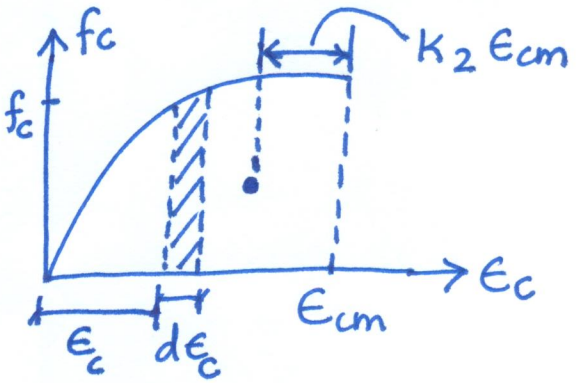
$$\int_0^{\epsilon_{cm}} f_c \, d\epsilon_c = K_1 K_3 f_c' \cdot \epsilon_{cm}$$

$$K_1 = \frac{\int_0^{\epsilon_{cm}} f_c \, d\epsilon_c}{K_3 f_c' \cdot \epsilon_{cm}}$$

Condition for K_2 :



$$K_2 = \frac{x}{Kd}$$



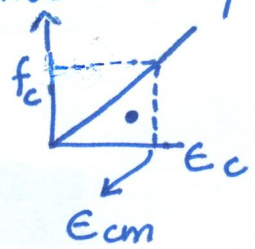
We take a differential area $f_c \, d\epsilon_c$ at a distance ϵ_c from origin.

$$\int_0^{\epsilon_{cm}} f_c \, d\epsilon_c \cdot \epsilon_c = \int_0^{\epsilon_{cm}} f_c \, d\epsilon_c \cdot (\epsilon_{cm} - K_2 \epsilon_{cm})$$

Sum of moments of all possible differential areas about origin = Moment of the total area (acting at its centroid)

Lets consider first, the most simplest case:

1) Linear $\sigma\epsilon$ model : ($K_3 = 1$)



$$K_1 = \frac{\text{Area of actual } \sigma\epsilon \text{ model}}{K_3 f_c' \cdot \epsilon_{cm}}$$

$$K_1 = \frac{\frac{1}{2}\epsilon_{cm} \cdot f_c}{f_c' \cdot \epsilon_{cm}} = \frac{1}{2}\frac{f_c}{f_c'} = 0.5\frac{f_c}{f_c'}$$

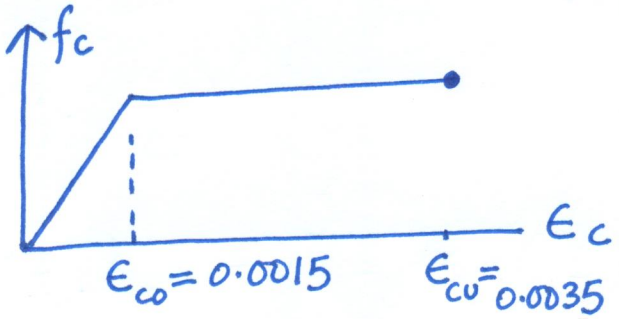
$$\frac{1}{2} f_c \epsilon_{cm} \cdot (\epsilon_{cm} - K_2 \epsilon_{cm}) = \int_0^{\epsilon_{cm}} f_c \epsilon_c \, d\epsilon_c$$

For linear elastic model, $f_c = E\epsilon_c$

So,
$$\frac{1}{2} \cdot E_c \cdot \epsilon_{cm}^2 (\epsilon_{cm} - K_2 \epsilon_{cm}) = E \int_0^{\epsilon_{cm}} \epsilon_c^2 \, d\epsilon_c$$

$$K_2 = \frac{1}{3}$$

2) Bilinear (Elastoplastic) $\sigma\epsilon$ Model :-



For $\epsilon_{cm} \leq \epsilon_{co}$,

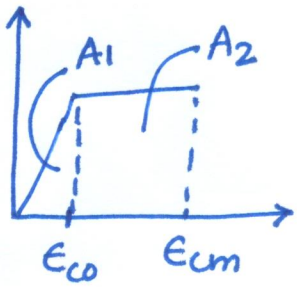
$$K_1 = 0.5\frac{f_c}{f_c'} = \frac{0.5 E_c \epsilon_{cm}}{E_c \epsilon_{co}} = 0.5\frac{\epsilon_{cm}}{\epsilon_{co}}$$

$$K_2 = \frac{1}{3}$$

For $0.0015 \le \epsilon_{cm} \le 0.0035$,

$$K_1 f_c' \epsilon_{cm} = \sum A_i$$

$$= A_1 + A_2$$



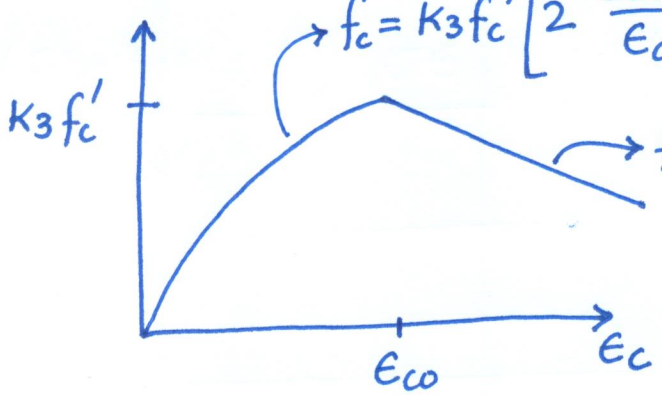
$$K_1 = 1 - \frac{\epsilon_o}{2\epsilon_{cm}}$$

$$\underbrace{\frac{1}{2} f_c' \epsilon_{co}}_{A_1} \cdot \underbrace{\frac{2}{3}\epsilon_{co}}_{\text{moment arm}} + \underbrace{f_c'(\epsilon_{cm} - \epsilon_{co})}_{A_2} \underbrace{\left(\epsilon_{co} + \frac{\epsilon_{cm} - \epsilon_{co}}{2}\right)}_{\text{arm}}$$

$$= \underbrace{(A_1 + A_2)}_{\int f_c d\epsilon_c \approx \sum A_i} \underbrace{(\epsilon_{cm} - K_2 \epsilon_{cm})}_{\text{moment arm of whole area under actual } \sigma\epsilon \text{ model}}$$

$$K_2 = \frac{\epsilon_{cm}^2 + \frac{1}{3}\epsilon_{co}^2 - \epsilon_{cm}\epsilon_o}{\epsilon_{cm}(2\epsilon_{cm} - \epsilon_o)}$$

3) Hognestad's Parabola with a Linear Tail



Z = Slope of Linear Tail

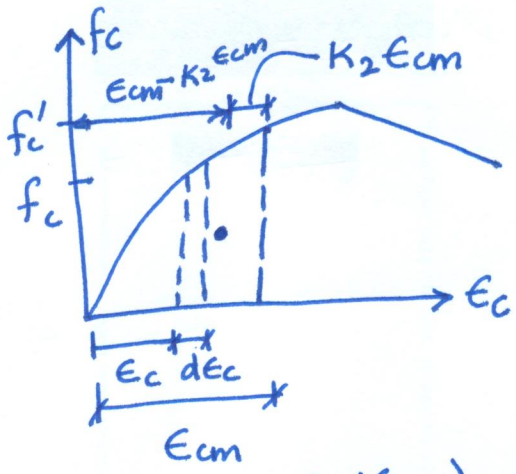
>250 for unconfined concrete.

For $0 \le \epsilon_{cm} \le \epsilon_{co}$

$$K_1 = \frac{1}{f_c' \epsilon_{cm}} \int_0^{\epsilon_{cm}} f_c \, d\epsilon_c = \frac{\epsilon_{cm}}{\epsilon_{co}} - \frac{1}{3}\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right)^2$$

For K_2,

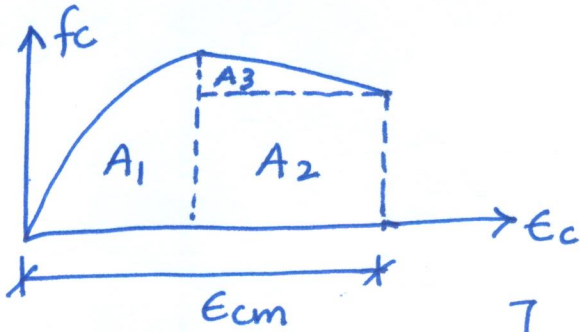
$$\int_0^{\epsilon_{cm}} f_c \, d\epsilon_c \; (\epsilon_{cm} - K_2 \epsilon_{cm}) = \int_0^{\epsilon_{cm}} f_c \epsilon_c \, d\epsilon_c$$



$$K_2 = 1 - \frac{\frac{2}{3}\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right) - \frac{1}{4}\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right)^2}{\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right) - \frac{1}{3}\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right)^2}$$

$$= 1 - \frac{\frac{1}{3} - \frac{1}{12}\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right)}{1 - \frac{1}{3}\left(\frac{\epsilon_{cm}}{\epsilon_{co}}\right)}$$

For $\epsilon_{cm} > \epsilon_{co}$,



$$K_1 = \frac{1}{\epsilon_{cm}}\left[\epsilon_{cm} - \frac{\epsilon_{co}}{3} - \frac{Z}{2}(\epsilon_{cm} - \epsilon_{co})^2\right]$$

$$K_2 = 1 - \frac{1}{C_3}\left[\frac{5}{12}\epsilon_{co} C_1 + C_4\left(C_1 + \frac{C_2}{2\epsilon_{cm}}\right) + C_3\left(C_1 + \frac{C_2}{2\epsilon_{cm}}\right)\right]$$

where, $C_1 = \frac{\epsilon_{co}}{\epsilon_{cm}}$, $C_2 = \epsilon_{cm} - \epsilon_{co}$, $C_3 = \frac{1}{2} Z C_2^2$

$C_4 = C_2 - Z C_2^2$, $C_5 = \epsilon_{cm} - \frac{1}{3}\epsilon_0 - C_3 = \sum A_i$